

Acquisition of Chess Knowledge in AlphaZero

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Learning from machines (AlphaZero)

- Most methods we have looked so far try to interpret algorithms trained on human-generated data and labels
 - These interpretations may resemble human-understandable representations only because they learned from such data
- Can we interpret what the algorithms has been learning through its self-play training process?



Learning from machines - three pronged acquisition

- Probe for concepts
 - How closely is AlphaZero's internal representation **related to** chess concepts humans have already created?
 - Detection of human concepts from network activations
- Study behavioral changes
 - How do changing representations give rise to changing behaviors?
 - Evolution of AlphaZero vs. human's strategy in openings
- Investigate activations directly
 - Unsupervised methods using non-negative matrix factorisation (NMF) and direct measure of covariance to discover concepts not represented by humans in the past

Interpretability

- Concept-based (post-hoc) interpretability
 - Network probing / important features / mechanistic understanding
 - Challenges: correlation not causal
- Explainability in reinforcement learning
 - Structural causal models / reward difference explanations
 - Identify interesting points in behavioral trajectories

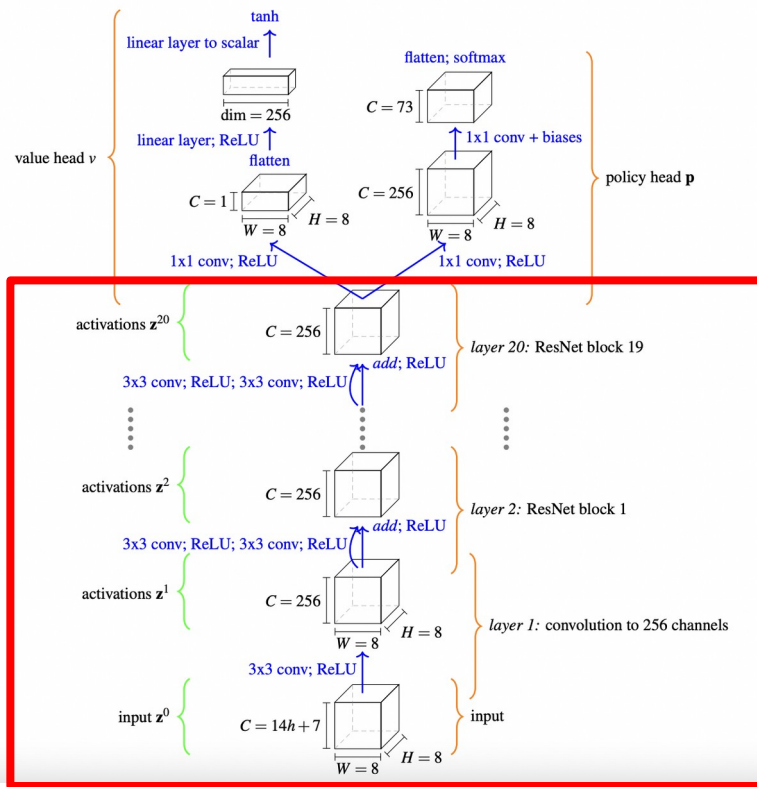
Chess as testing ground for AI interpretability

Can human learn the machine's strategy?

- Does not rely on human-labeled data
- Tree-based organization / saliency maps
- Natural language processing to generate move-by-move commentary
- **This paper:** captures “intuitive” aspect of chess play by understanding networks that produce value assessment (v) and candidate move ($\mathbf{p}$)

$$\mathbf{p}, v = f_{\boldsymbol{\theta}}(\mathbf{z}^0)$$

AlphaZero: Network structure and training



$$\mathbf{p}, v = f_{\theta}(\mathbf{z}^0)$$

$$\mathbf{z}^l = f^l(\mathbf{z}^{l-1}) = \text{ReLU}(\mathbf{z}^{l-1} + g^l(\mathbf{z}^{l-1}))$$

$$\mathbf{z}^l = f^{1:l}(\mathbf{z}^0) = f^l \circ \dots \circ f^2 \circ f^1(\mathbf{z}^0)$$

Probing for concepts

Encoding of human conceptual knowledge

Question: Can human concepts be easily predicted from network's internal representation?

Concept: User defined function mapping network input to real line.

$$c(\mathbf{z}^0) = \begin{cases} 1 & \text{if } \mathbf{z}^0 \text{ contains a } \text{♔}\text{-pair for the playing side} \\ 0 & \text{otherwise} \end{cases}$$

Approach: Train a sparse linear regression model from activations $\mathbf{z}^l$ at layer l and training step t to human concept j

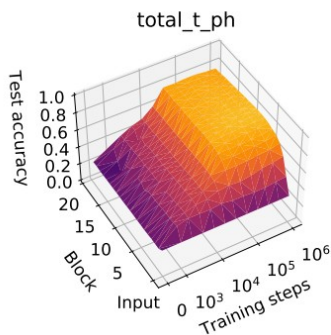
Probing concept learning: details

- **Data:** Randomly sample training, validation, test data from ChessBase archive, compute concept values and AlphaZero activations
- **Procedure:** For concept i . layer l . training step t . solve

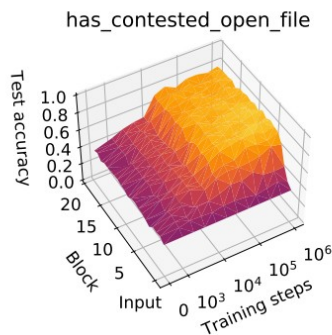
$$\mathbf{w}_{jlt}, b_{jlt} = \min_{\mathbf{w}, b} \frac{1}{N} \left\| \mathbf{w}^T \mathbf{Z}_t^l + b \mathbf{1} - \mathbf{c}_j \right\|_2^2 + \lambda \|\mathbf{w}\|_1 + \lambda |b|$$

- **Controls:** Regression from $\mathbf{z}^0$ and random concept regression.
- **Evaluation:** R^2 value, fraction of variance in concept explained by network activation

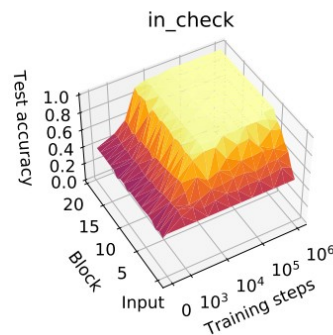
Evolution of human concepts in AlphaZero



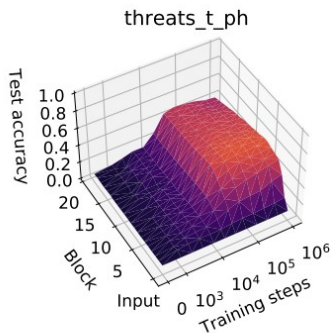
(a) Stockfish 8's total score



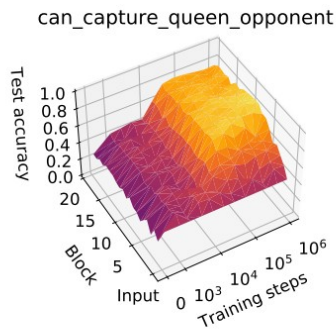
(b) A contested open file is occupied by rooks and/or queens of opposite colours



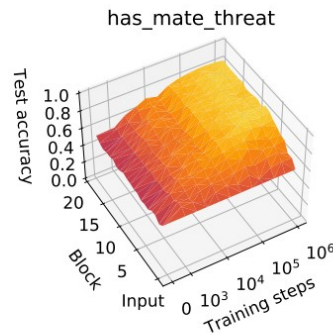
(c) Is the playing side in check?



(d) Stockfish 8's evaluation of threats.



(e) Can the playing side capture their opponent's queen?



(f) Could the opposing side checkmate the playing side in one move?

Key findings

- 1) **Grokking:** Many concepts begin to increase in accuracy around 32,000 steps
- 2) Drop in linearly-available information in later layers for some concepts
- 3) Some concepts **cannot** be regressed: sparsity partially obstructs ability to relate **highly-distributed representations** to concepts
- 4) **Learning from prediction errors:** regression errors may point to a “difference of opinion” with Stockfish

Challenges for concept probing

- 1) What's the right **probing architecture**?
- 2) How should we interpret **complex or subjective concepts**?
- 3) When can we definitively say a **concept is represented**?
- 4) When we train a probe, we cannot tell if we are getting a **confounder or the concept** itself

Progression through AlphaZero and human history

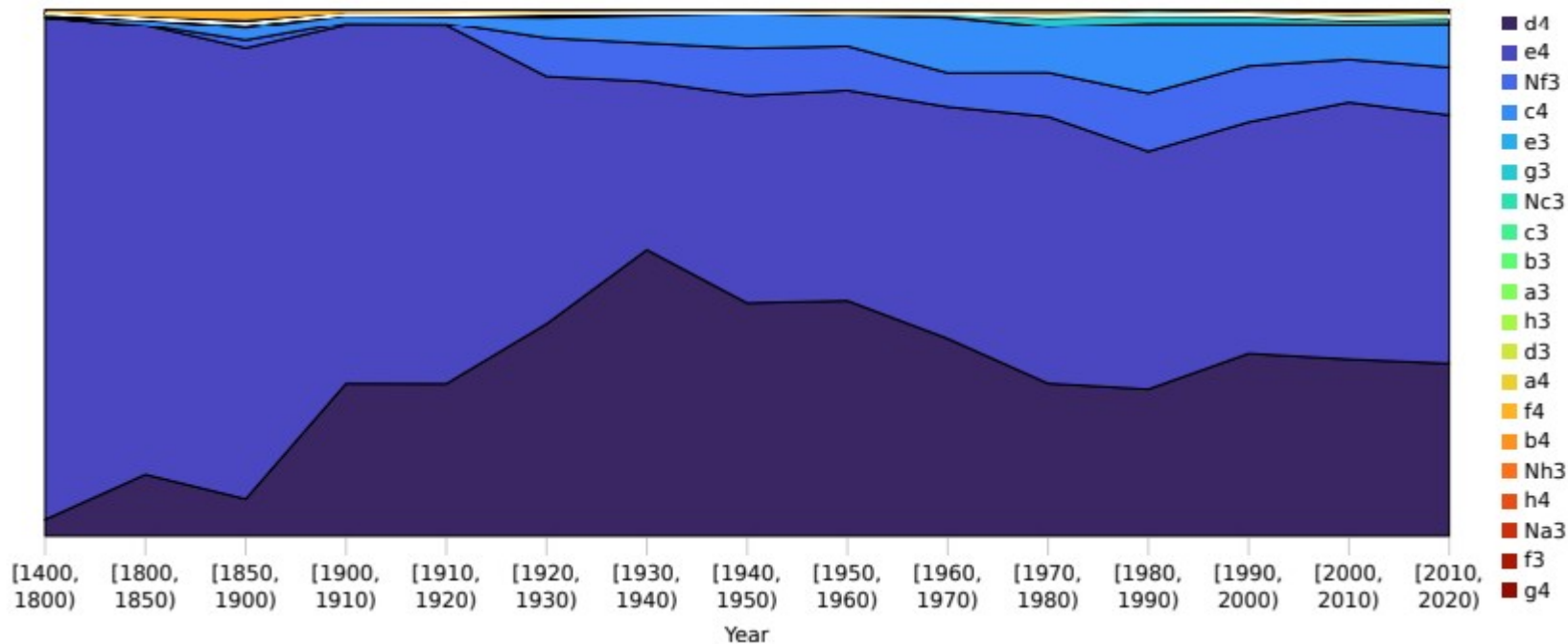
Progression of knowledge

- Recall:

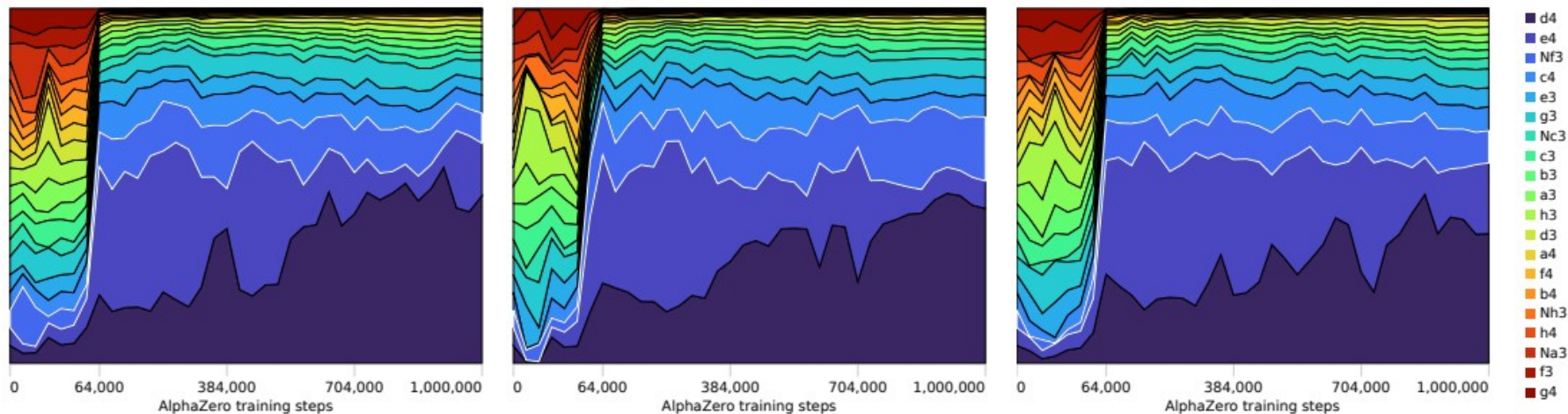
$$\mathbf{p}, v = f_{\boldsymbol{\theta}}(\mathbf{z}^0)$$

- **Question:** How does the progression of AlphaZero's knowledge compare to that of humans?
- **Key Findings:**
 - AlphaZero: Start from a uniform prior, then narrow down.
 - Humans: Start from a concentrated prior, then expand.

Progression through human history



Progression through AlphaZero history



Progression of AlphaZero's Chess Knowledge

Methodology

- ❖ At different training steps (up to 128k), examine:
 - AlphaZero's move tendencies
 - AlphaZero's concept encodings

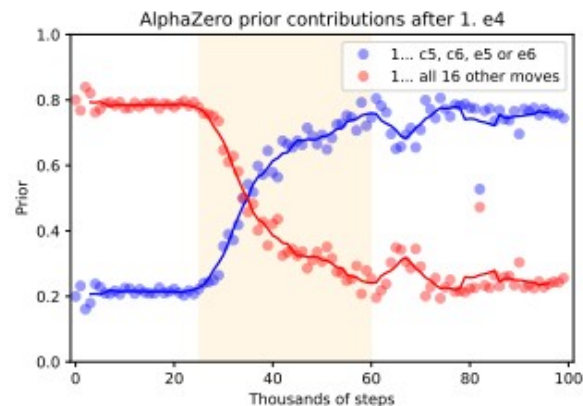
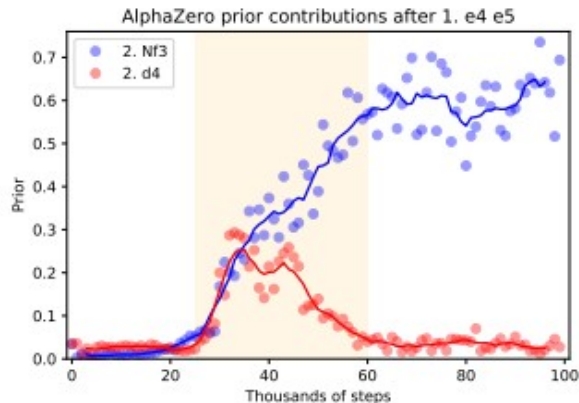
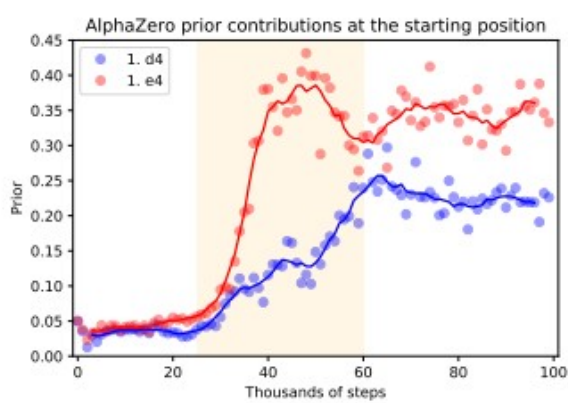
Primary Takeaways

- 1) AlphaZero's learns standard opening theory early on
- 2) AlphaZero learns material values before more complex positional concepts

Both reinforce idea that AlphaZero learns basic human chess concepts first

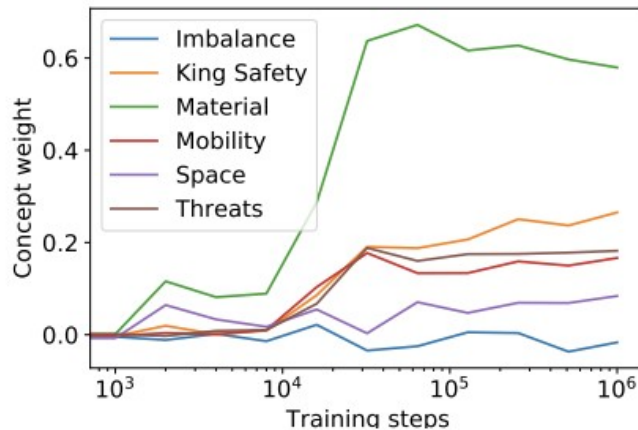
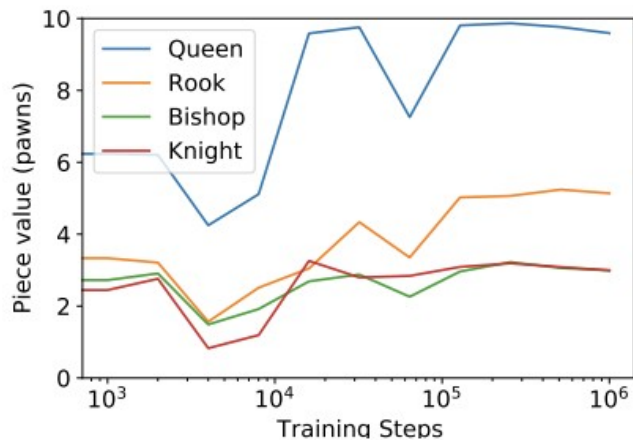
Opening Theory Knowledge

- 1) ~30-60k: AlphaZero plays 1. e4/d4 the majority of the time
- 2) ~45k: AlphaZero considers 2. d4 before opting for 2. Nf3
- 3) ~45k: AlphaZero plays standard responses to 1. e4



Material vs. Positional Knowledge

- 1) ~30k training steps: Piece Values develop, converge ~100k
- 2) King Safety, Mobility concepts emerge after Material
 - a) More complex concepts require more training time



$$\hat{v}_{\mathbf{w},b}(\mathbf{z}^0) = \tanh(\mathbf{w}^T \mathbf{c}(\mathbf{z}^0) + b)$$

$$\mathbf{w}_t, b_t = \min_{\mathbf{w},b} \frac{1}{N} \sum_n \left| \hat{v}_{\mathbf{w},b}(\mathbf{z}_n^0) - v_{\theta_t}(\mathbf{z}_n^0) \right|$$

Training Progression Assessment: GM Vladimir Kramnik

- 1) **16k to 32k:** Material Value in Complex Positions
 - 2) **32k to 64k:** King Safety in Imbalanced Positions
 - 3) **64k to 128k:** King Safety & Material Sacrifices
in Complex Positions
- ❖ Tactical skills appear to precede positional skills as AlphaZero learns



Exploring Additional Feature Detectors in AlphaZero Network

Exploring Activations with Unsupervised Methods

1) **Goal:** Find Feature Detectors embedded within Network

2) **Methods:**

- a) Non-Negative Matrix Factorization of each layer's channels
- b) Correlation of Input Board with each channel's activations

Primary Takeaway

- ❖ Individual network layers & channels encode *feature detectors* related to human-recognizable chess concepts

Approach #1: NN Matrix Factorization Analysis

For each layer l with C channels,

- 1) Compute a matrix factorization $\mathbf{\Omega} \mathbf{x} \mathbf{F}$ using $K < C$ columns

For each factor k ($1 \dots K$) and input n ($1 \dots N$)

- 2) Visualize activations on Chess Board to find *feature detectors*

$$\hat{\mathbf{Z}}^l \in \mathbb{R}^{NHW \times C}$$

$$\mathbf{\Omega}_{\text{all}} \in \mathbb{R}^{NHW \times K}$$

$$\mathbf{F} \in \mathbb{R}^{K \times C}$$

$$\mathbf{F}^*, \mathbf{\Omega}_{\text{all}}^* = \min_{\mathbf{F}, \mathbf{\Omega}_{\text{all}}} \|\hat{\mathbf{Z}}^l - \mathbf{\Omega}_{\text{all}} \mathbf{F}\|_2^2$$

$$\mathbf{F}, \mathbf{\Omega}_{\text{all}} \geq \mathbf{0} .$$

Results: NN Matrix Factorization Analysis



Diagonal Moves
(1st Layer)



Diagonal Moves
(2nd Layer)



of Pieces that could
move to each square

Approach #2: Input-Activation Covariance Analysis

For each layer l and channel i ,

- 1) Compute the covariance between input $\mathbf{z}^0$ and position activations
- 2) Visualize covariances on Chess Board to find *feature detectors*

$$\text{cov}(z_i^l, \mathbf{z}^0) = \mathbb{E}[z_i^l \mathbf{z}^0] - \mathbb{E}[z_i^l] \mathbb{E}[\mathbf{z}^0]$$

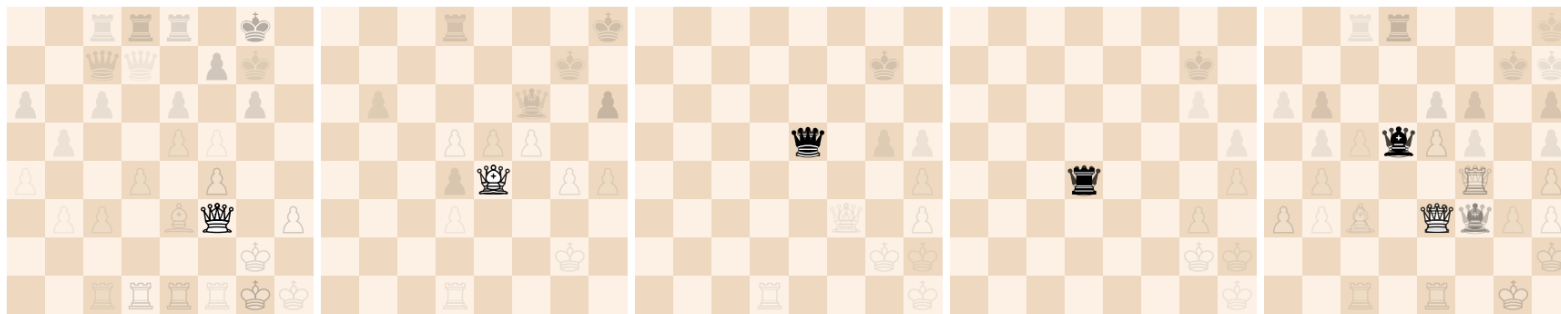
Results: Input-Activation Covariance Analysis

Detecting Move-Types from a Square

1-2) Diagonal-Attacking Pieces (Queen, Bishop)

3-4) Horizontally-Attacking Pieces (Queen, Rook)

5) Both



5 covariances with different channels for the square (5, 4)

Conclusion - Key Findings

- 1) Human-Defined Concepts can be regressed from the AlphaZero network
 - a) Despite never being trained on a human game of chess
- 2) As training progresses, AlphaZero understands basic concepts (openings, material) before more complex ones (king safety, mobility)
- 3) Feature Detectors of human-recognizable chess concepts are encoded by individual layers + channels in AZ Network
 - a) Can be found using supervised & unsupervised techniques

Limitations

- 1) Challenges for Concept Probing (previous slide)
- 2) Knowledge acquisition is not complete - only a very small part of the model
- 3) Interpreting “Feature Detectors” found via Unsupervised techniques
 - a) Inherently subjective
- 4) No Causal Insight into the Learned Concepts (only correlation)

Future Research Areas

- 1) Addressing Concept Probing Limitations (more than sparse LR)
- 2) Can we go beyond finding human knowledge embedded in AlphaZero and understand what new concepts are learned?
 - a) Further analysis of feature detectors found with unsupervised techniques
- 3) How do we generalize these findings to other machine learning settings?
Could we find human-recognizable concepts in different trained models?

Discussion

- How is this paper different from methods we discussed so far?
 - In terms of goals / techniques / specific models to explain
- Does this approach (or anything that can be built up on this) have potential applications for a more general setting other than board games?
 - What are the pros/cons of using games/artificial environments as baseline for evaluating RL algorithms?
- Can this be applied to our practice of doing science?
 - Can we recreate or extract theories using similar frameworks (AI for science)?
- How should we define or understand “knowledge” in these settings?
 - How is it different from similar to concepts/models/explanations/information?
- How convinced are you convinced about the “interpretability” aspect?
 - Who would be potential audience that could benefit from such analysis?

What the DAAM: Interpreting Stable Diffusion Using Cross Attention

Authors: Raphael Tang, Linqing Liu, Akshat Pandey, Zhiying Jiang, Gefei Yang, Karun Kumar, Pontus Stenetorp, Jimmy Lin, Ferhan Ture

Presented by: Alex Lin, Jason Jabbour, Mark Mazumder

Outline

- Related work
- Background: (simplified) Stable Diffusion
- **DAAM** for text-to-image **attribution**
- Results
 - Attribution quality analysis
 - Syntax to pixels
 - Entanglement
- Limitations & discussion

DAAM estimates per-pixel attribution for each word in a prompt (post-hoc)

Stable Diffusion: Text to Image

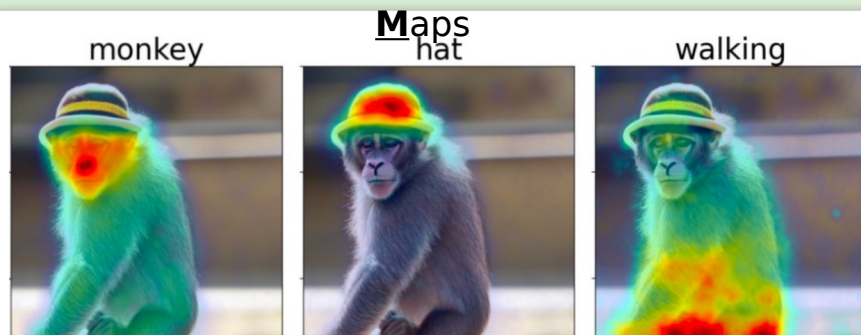
INPUT TEXT PROMPT

“monkey with hat walking”

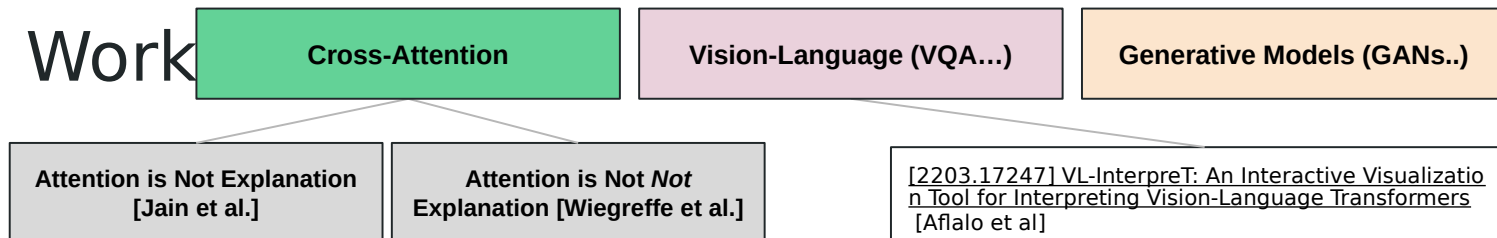
OUTPUT IMAGE



DAAM: Diffusion Attentive Atribution



Related Work

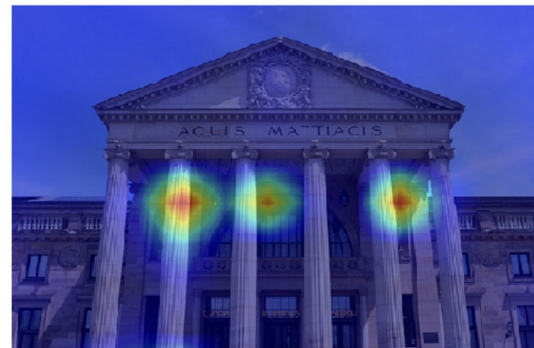


This work applies existing techniques (cross-attention) to an **open source, SOTA** diffusion model to probe limitations

- **Textual perturbation** (Wallace et al., 2019), Attentional Visualization (Vig, 2019; Kovaleva et al., 2019, Shimoaka et al., 2016) and information bottlenecks (Jiang et al., 2020) to relate **important input tokens** to **outputs of large language models**
- Probing **vision transformers for verb understanding** (Hendricks and Nematzadeh, 2021)
- Enhancing diffusion models using **prompt engineering** (Hertz et al, 2020; Woolf, 2020)
- **Disentangling** e.g., style and spelling (Karras et al., 2019; Materzynska et al., 2022)

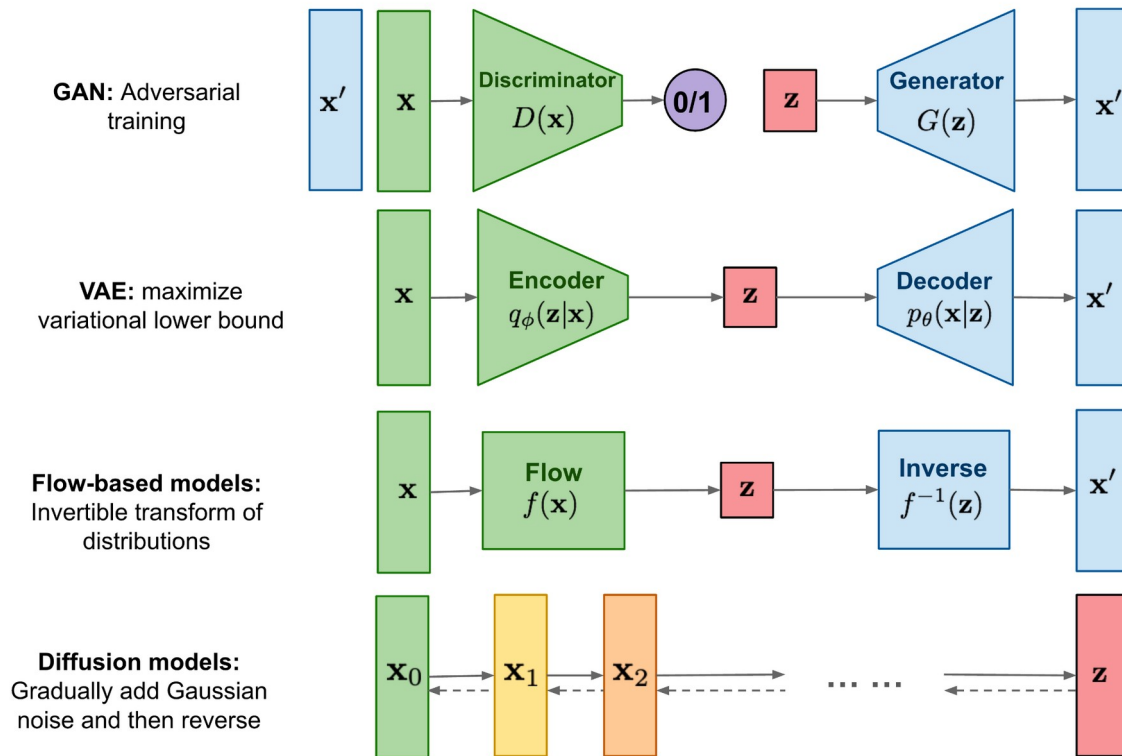
[2203.17247] VL-InterpreT: An Interactive Visualization Tool for Interpreting Vision-Language Transformers [Aflalo et al]

How many pillars are in front of the Façade of the Kurhaus ?



(b) **Predicted:** *There are 6 pillars in front of the Façade of the Kurhaus.*

Diffusion Models vs: GANs, VAEs, Flow

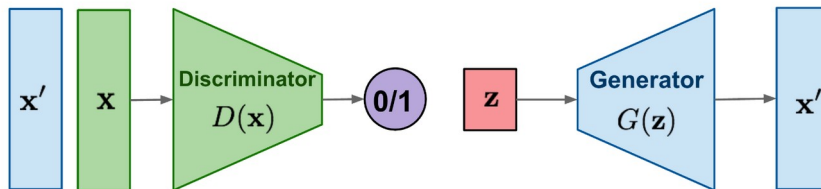


Source: <https://lilianweng.github.io/posts/2021-07-11-diffusion-models/>

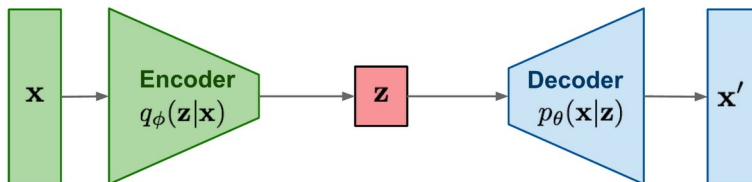
Diffusion Models vs: GANs, VAEs, Flow

“single-pass”
encoder-
decoder models

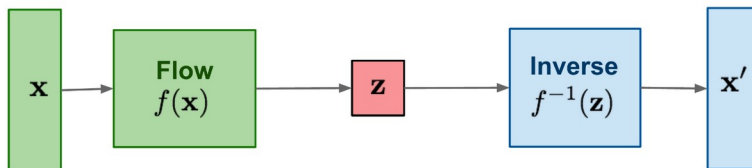
GAN: Adversarial
training



VAE: maximize
variational lower bound



Flow-based models:
Invertible transform of
distributions



Diffusion models:
Gradually add Gaussian
noise and then reverse

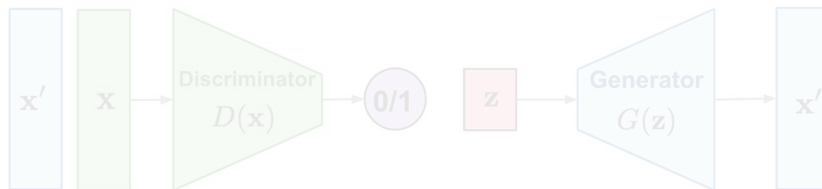


Source: <https://lilianweng.github.io/posts/2021-07-11-diffusion-models/>

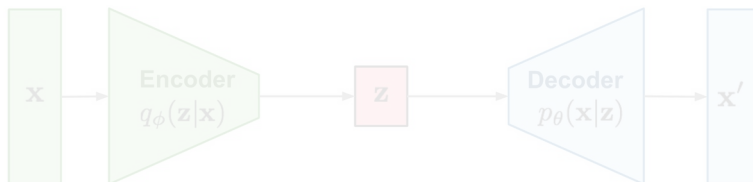
Diffusion Models vs: GANs, VAEs, Flow

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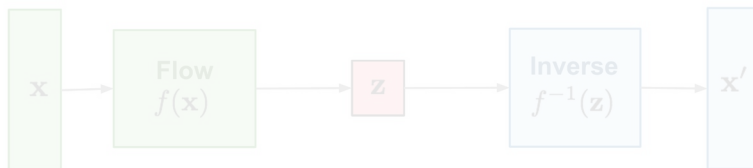
GAN: Adversarial
training



VAE: maximize
variational lower bound

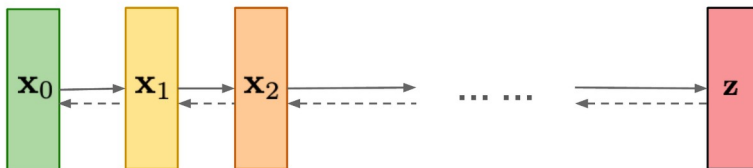


Flow-based models:
Invertible transform of
distributions



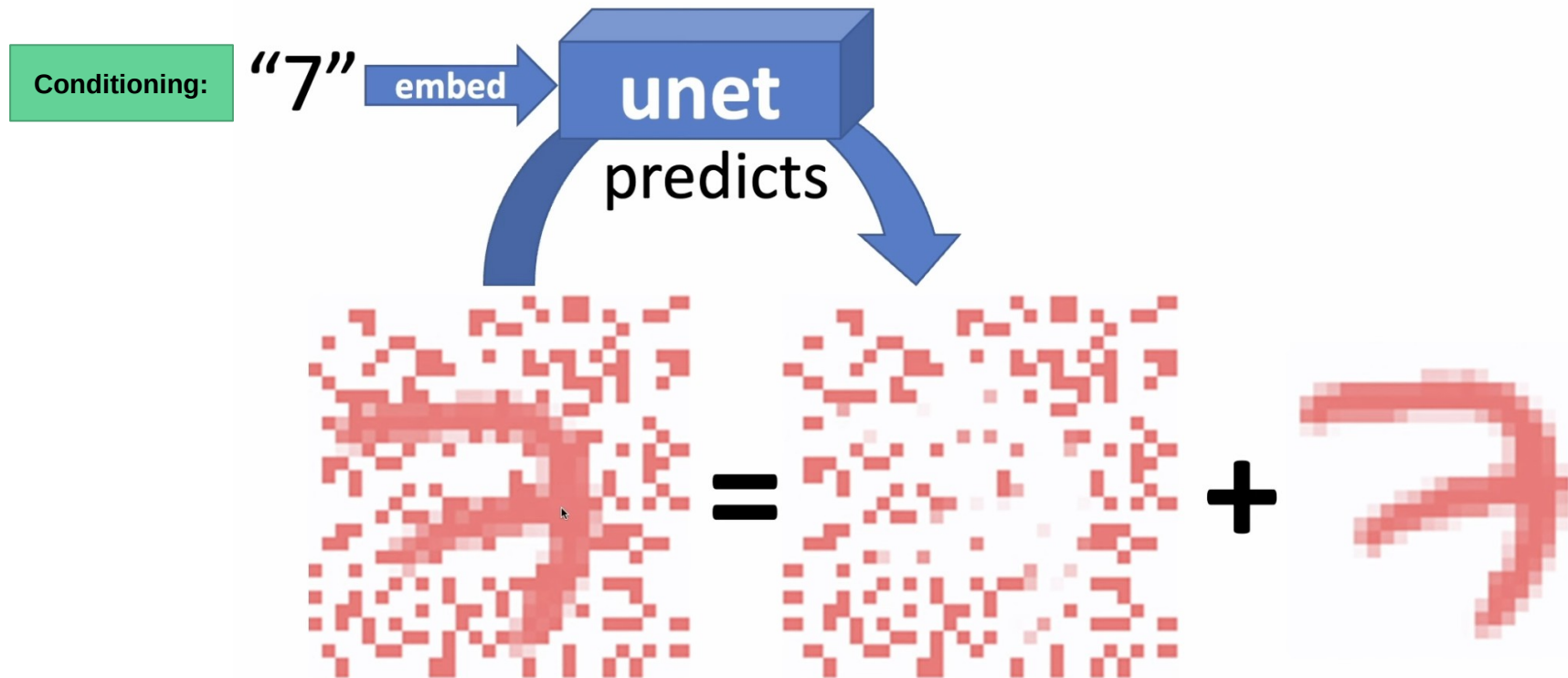
**Multi-step
time-series**

Diffusion models:
Gradually add Gaussian
noise and then reverse



Source: <https://lilianweng.github.io/posts/2021-07-11-diffusion-models/>

Image Generation: Predict and subtract noise

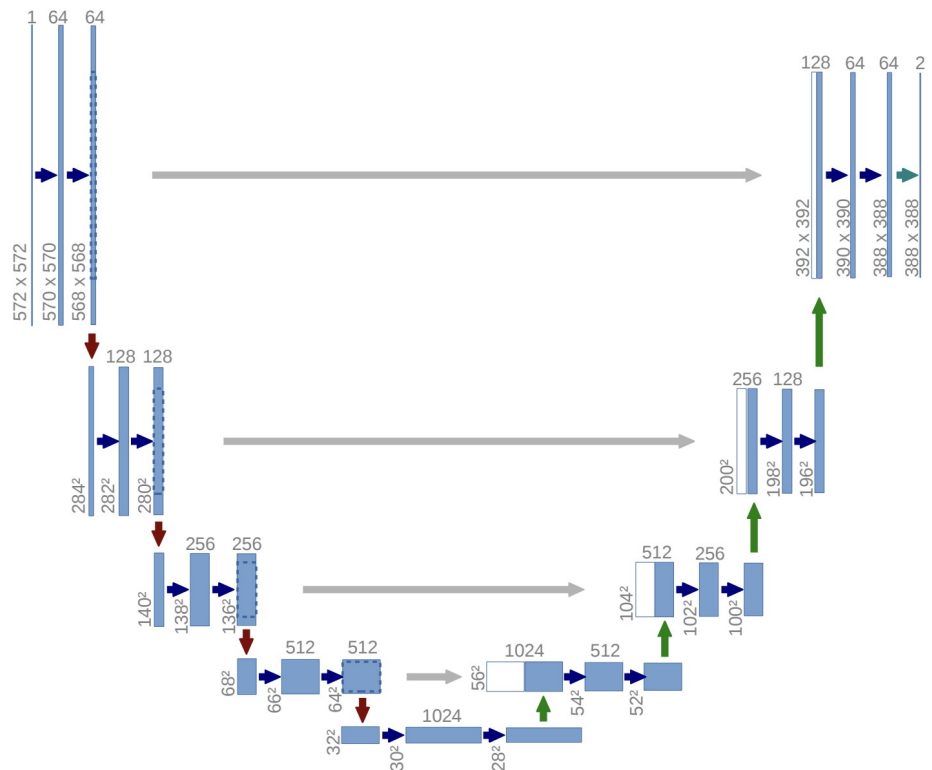


Source:

[Lesson 10: Deep Learning Foundations to Stable Diffusion, 2022](#)

UNet Architecture

1505.04597] U-Net: Convolutional Networks for Biomedical Image Segmentation



Forward Process: Noise Schedule

Fixed additive noise model: ϵ

Add noise for time steps t in $\{0, \dots, 30\}$



...

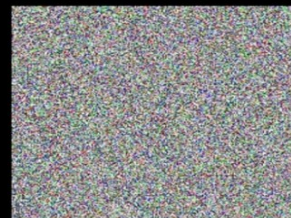


Image Source: [Diffusion models from scratch in PyTorch](#)

Reverse Process: Learned Noise Estimate

Fixed additive noise model: ϵ

Add noise for time steps t in $\{0, \dots, 30\}$



...

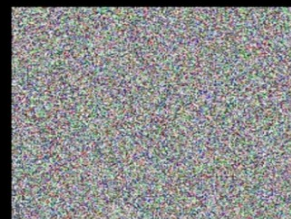
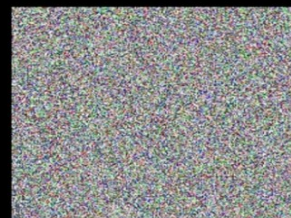


Image Source: [Diffusion models from scratch in PyTorch](#)

Denoising model: $\epsilon_{\theta}(\text{image}, \text{timestep})$



...



Diffusion Model Loss

$$L_{DM} = \mathbb{E}_{x, \epsilon \sim \mathcal{N}(0,1), t} \left[\|\epsilon - \epsilon_{\theta}(x_t, t)\|_2^2 \right]$$

Error of **noise
reconstruction**

Diffusion Model Loss

$$L_{DM} = \mathbb{E}_{x, \epsilon \sim \mathcal{N}(0,1), t} \left[\|\epsilon - \epsilon_{\theta}(x_t, t)\|_2^2 \right]$$

Error of **noise reconstruction**

Added noise (from Normal distribution)

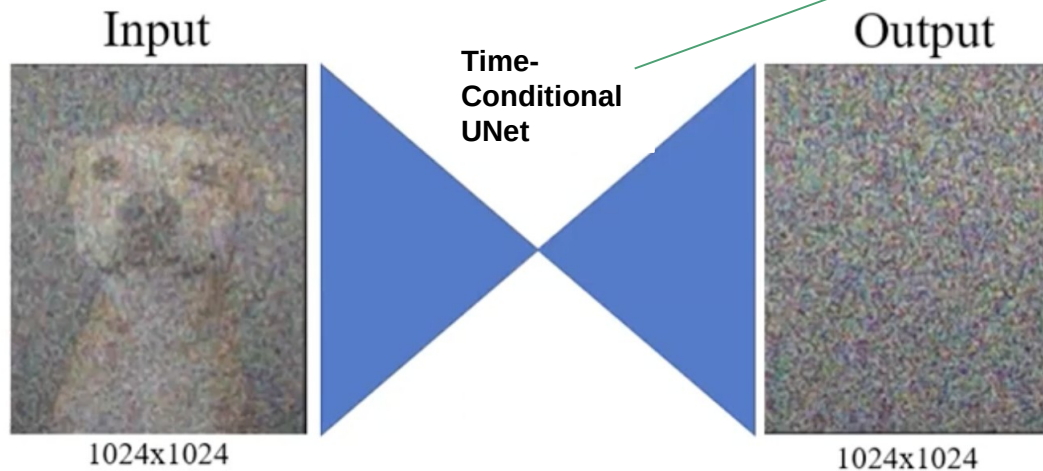
Denoising Model

Squared Euclidean Norm

Image at time t

Diffusion Model Loss

$$L_{DM} = \mathbb{E}_{x, \epsilon \sim \mathcal{N}(0,1), t} \left[\|\epsilon - \epsilon_{\theta}(x_t, t)\|_2^2 \right]$$

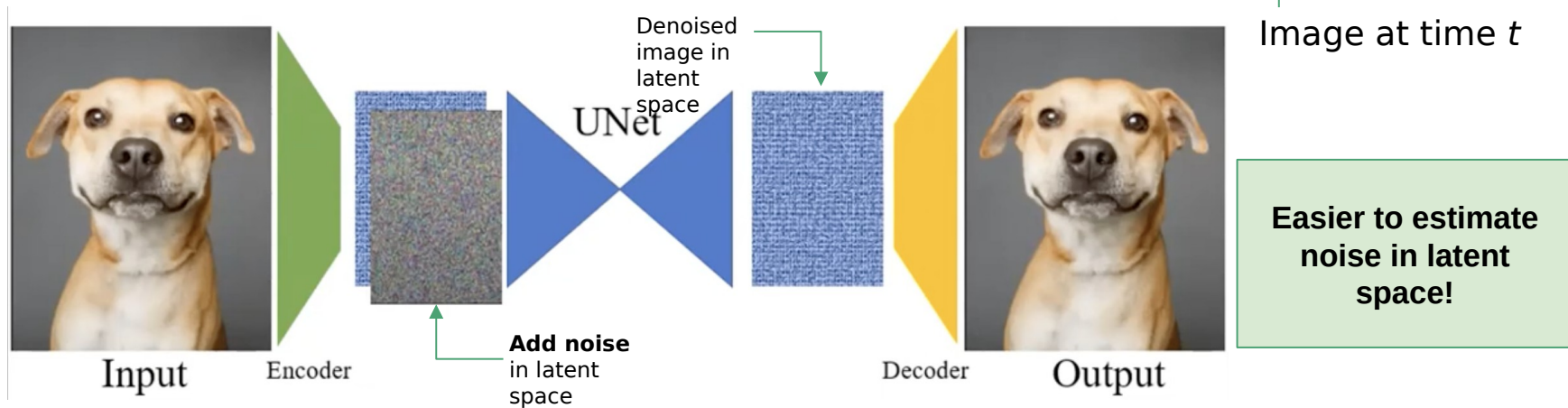


Reasoning about noise in pixel space is challenging

Source: [High-Resolution Image Synthesis with Latent Diffusion Models](#)

Diffusion Model Loss

$$L_{DM} = \mathbb{E}_{x, \epsilon \sim \mathcal{N}(0,1), t} \left[\|\epsilon - \epsilon_{\theta}(x_t, t)\|_2^2 \right]$$



Source: [High-Resolution Image Synthesis with Latent Diffusion Models](#)

Latent Diffusion Model Loss

Added noise (from Normal distribution)

Denoising Model

Squared Euclidean Norm

$$L_{DM} = \mathbb{E}_{x, \epsilon \sim \mathcal{N}(0,1), t} \left[\|\epsilon - \epsilon_{\theta}(x_t, t)\|_2^2 \right]$$

Image at time t

$$L_{LDM} := \mathbb{E}_{\underbrace{\mathcal{E}(x)}_{\text{Latent space encoder}}, \epsilon \sim \mathcal{N}(0,1), t} \left[\|\epsilon - \epsilon_{\theta}(z_t, t)\|_2^2 \right]$$

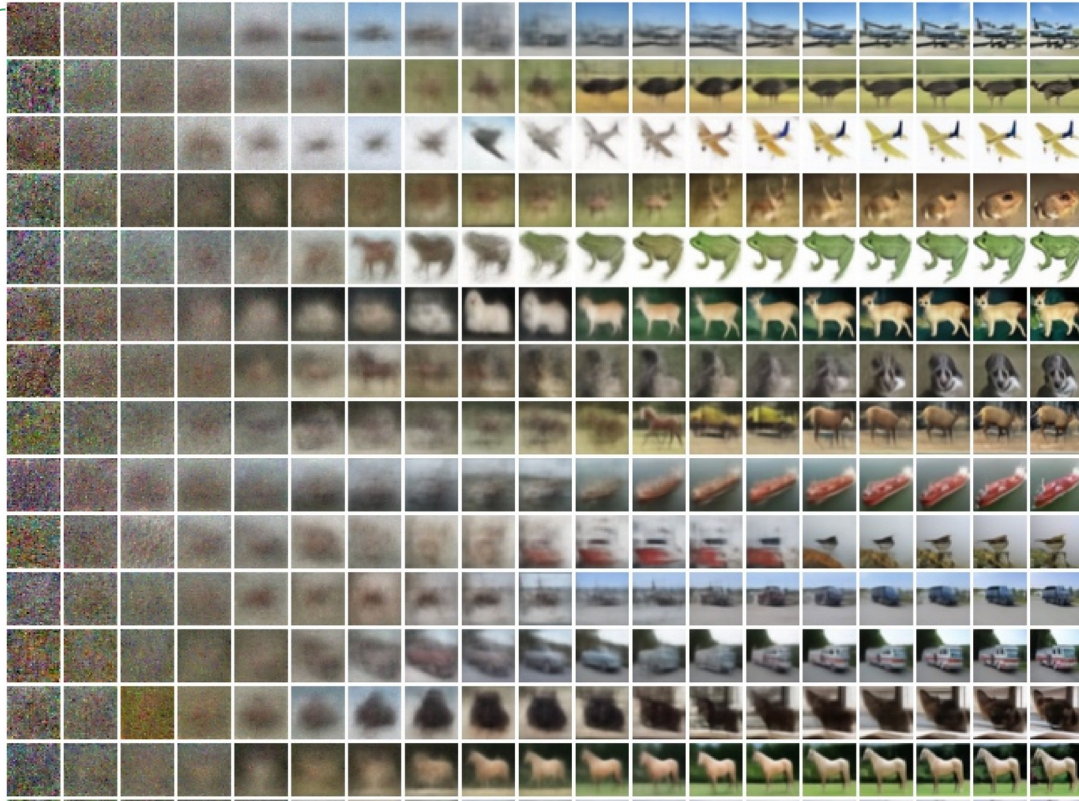
Latent representation of image at time t

Source: [High-Resolution Image Synthesis with Latent Diffusion Models](#)

Unconditional Sampling from Noise

<https://arxiv.org/abs/2006.11239>

Random
Noise

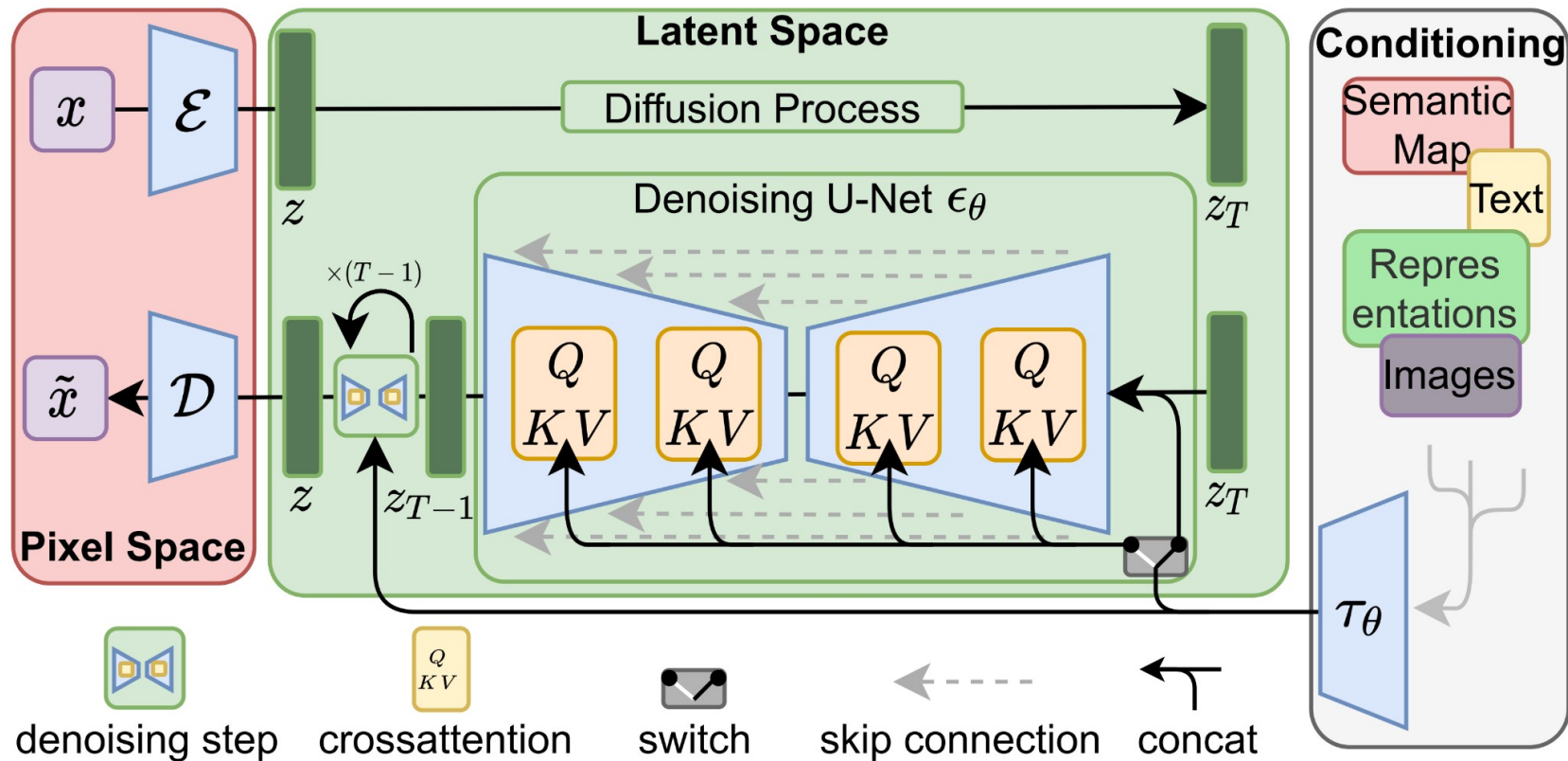


Conditional Generation?

How do we go from a
text prompt to an
image?

Stable Diffusion Architecture

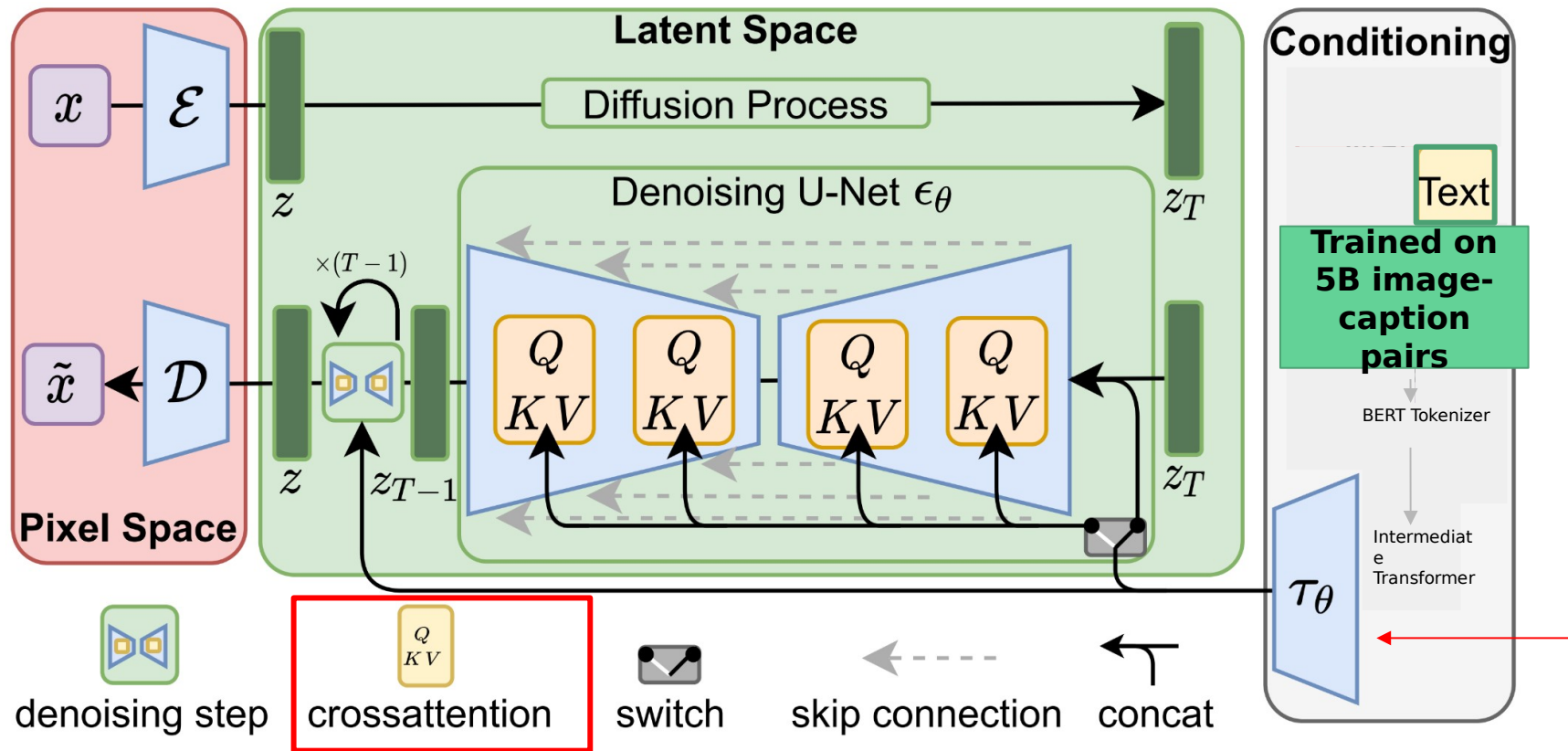
arXiv:2212.097521 High-Resolution Image Synthesis with Latent Diffusion Models



$$L_{LDM} := \mathbb{E}_{\mathcal{E}(x), y, \epsilon \sim \mathcal{N}(0,1), t} \left[\left\| \epsilon - \underbrace{\epsilon_{\theta}(z_t, t, \tau_{\theta}(y))}_{\text{Additionally parameterized on text encoding}} \right\|_2^2 \right]$$

Stable Diffusion Architecture

arXiv:2212.07521 High-Resolution Image Synthesis with Latent Diffusion Models



Cross-Attention for Text Conditioning

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d}} \right) \cdot V$$

UNet block i

Text encoder

Text encoder

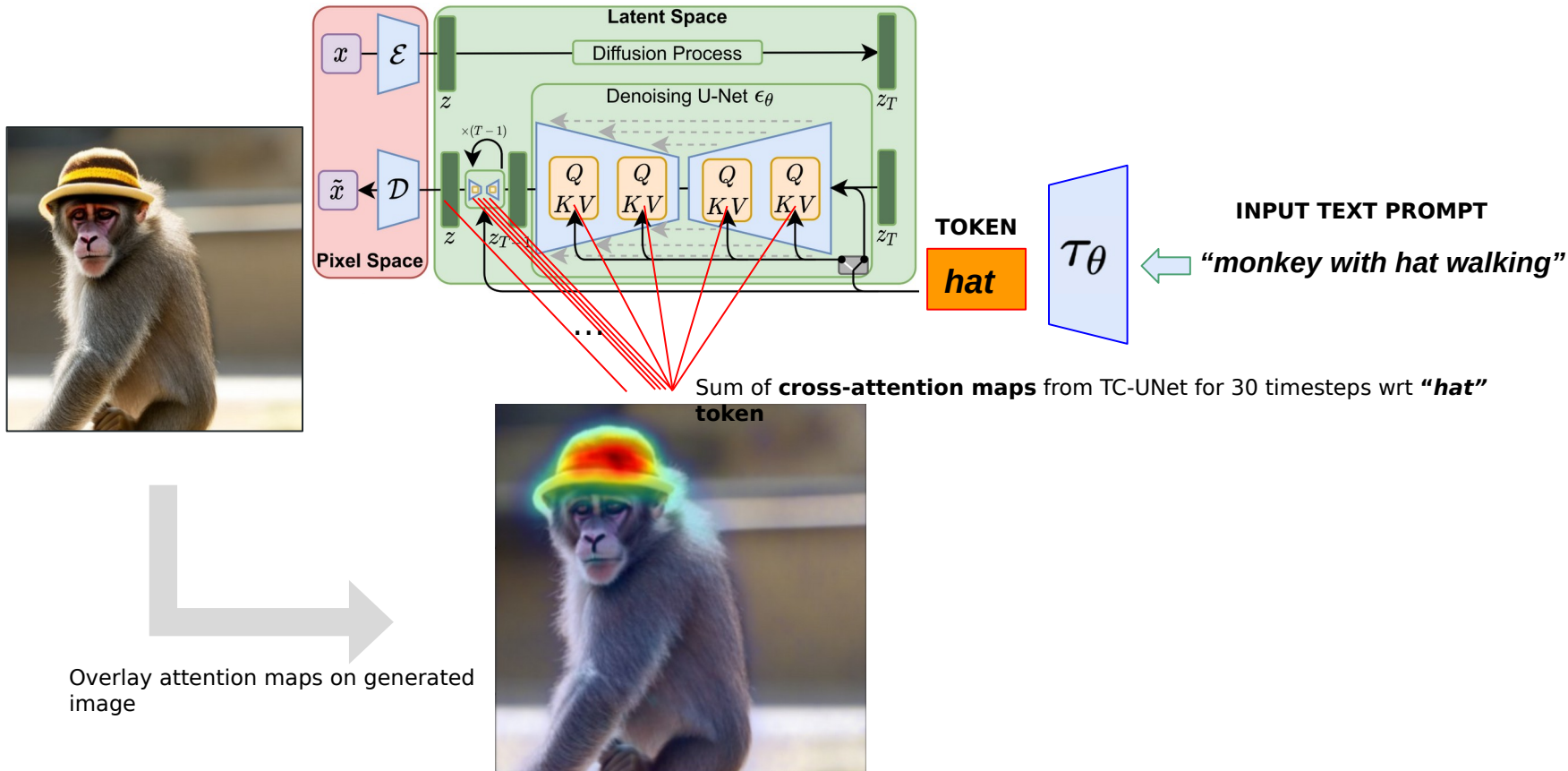
$$Q = W_Q^{(i)} \cdot \varphi_i(z_t), \quad K = W_K^{(i)} \cdot \tau_\theta(y), \quad V = W_V^{(i)} \cdot \tau_\theta(y)$$

Queries: image tokens

Keys, Values: prompt tokens

DAAM estimates **per-word attribution** via this cross-attention **for each subset** of prompt tokens

DAAM Per-Word Heatmaps



DAAM Attention Maps

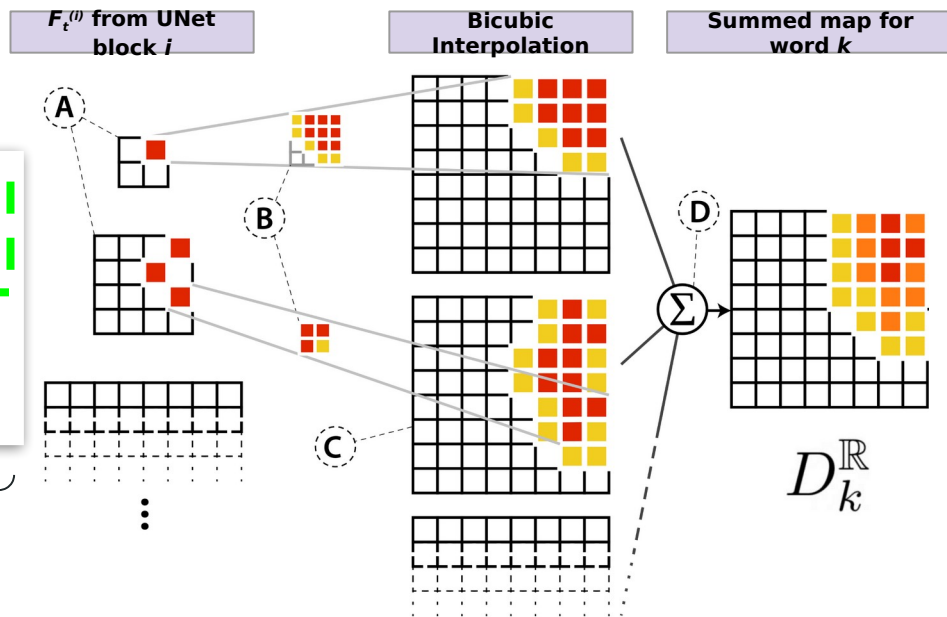
$$F_t^{(i)\downarrow} \in \mathbb{R}^{\left\lceil \frac{w}{c^i} \right\rceil \times \left\lceil \frac{h}{c^i} \right\rceil \times l_H \times l_W}$$

Cross-Attention Map
for block i at time t

Width, height of
downsampled
UNet block

Number of
attention heads

Number of
words in
prompt



$$D_k^{\mathbb{R}}[x, y] := \sum_{i, j, \ell} \tilde{F}_{t_j, k, \ell}^{(i)\downarrow}[x, y] + \tilde{F}_{t_j, k, \ell}^{(i)\uparrow}[x, y]$$

Summed map for each
word k , $k \in \{1, \dots, l_W\}$

i, j, ℓ

Downsampling
blocks

Upsampling blocks

Sum across: UNet blocks i , timesteps j , attention
heads ℓ

Outline

- Background: Stable Diffusion
- DAAM for text-to-image attribution
- Results
 - Attribution quality analysis
 - Syntax to pixels
 - Entanglement
- Related work
- Limitations & discussion

Attribution Analysis Part 1: Object Attribution

We can evaluate DAAM as an image-segmentation tool



DAAM segments Stable Diffusion images

# Method	COCO-Gen		Unreal-Gen	
	mIoU ⁸⁰	mIoU [∞]	mIoU ⁸⁰	mIoU [∞]
Supervised Methods				
1 Mask R-CNN (ResNet-101)	82.9	32.1	76.4	31.2
2 QueryInst (ResNet-101-FPN)	80.8	31.3	78.3	35.0
3 Mask2Former (Swin-S)	84.0	32.5	80.0	36.7
4 CLIPSeg	78.6	71.6	74.6	70.9
Unsupervised Methods				
5 Whole image mask	20.4	21.1	19.5	19.3
6 PiCIE + H	31.3	25.2	34.9	27.8
7 STEGO (DINO ViT-B)	35.8	53.6	42.9	54.5
8 Our DAAM-0.3	64.7	59.1	59.1	58.9
9 Our DAAM-0.4	64.8	60.7	60.8	58.3
10 Our DAAM-0.5	59.0	55.4	57.9	52.5


$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$

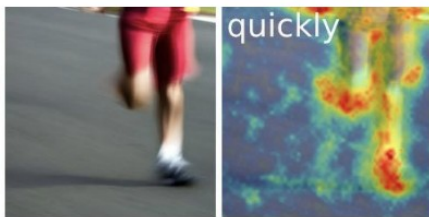

- DAAM does not require explicit segmentation labels
- DAAM is “open vocabulary” – can segment for any text input (not limited to known classes)

Attribution Analysis Part 2: Generalized Attribution

DAAM can segment beyond nouns



NUM



ADV



ADJ



VERB

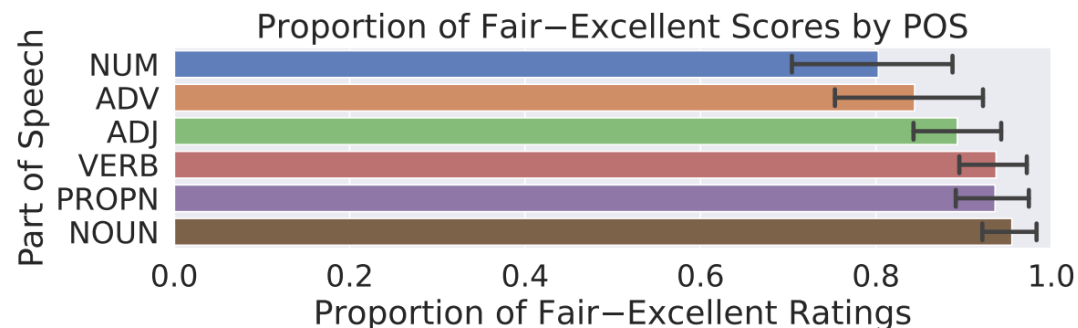
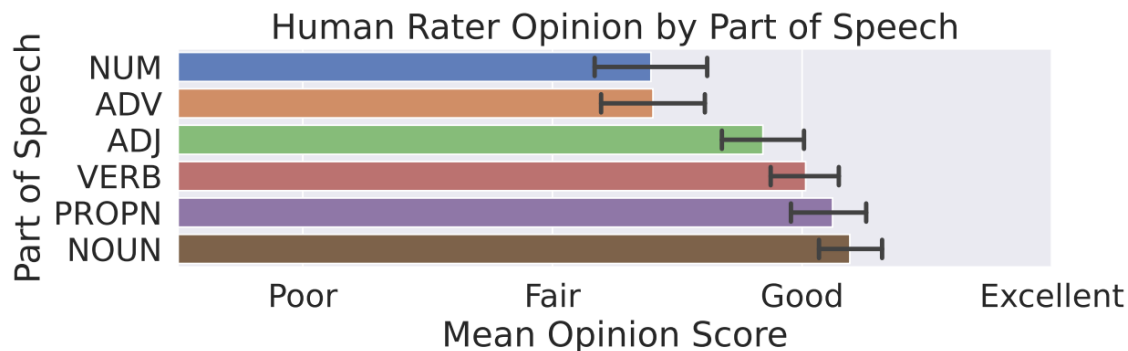


PROPN

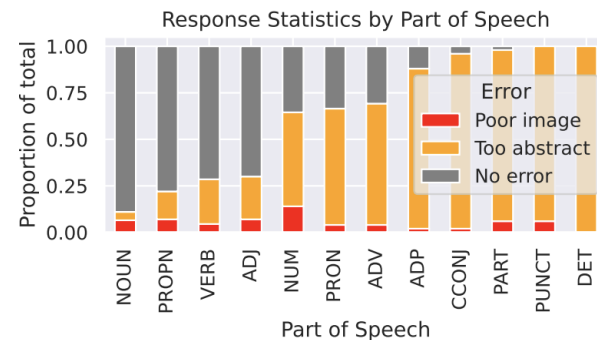


NOUN

Evaluating DAAM with a user study



- 50 annotators, none see more than 18% of images
- Every image has three raters
- Abstract words / poor images were thrown out of evaluation



Visuosyntatic Analysis



How does **syntax** relate to the generated **pixels**?

Visuosyntatic Analysis

Head-Dependent Pairs

The cat sat on the mat

The red car

She ate the pizza

Visuosyntactic Analysis

Head-Dependent Pairs

The cat sat on the mat

Head because subject of sentence

Dependent because mat is the object of the sentence

The red car

Dependent because "red" is an adjective that describes the noun "car"

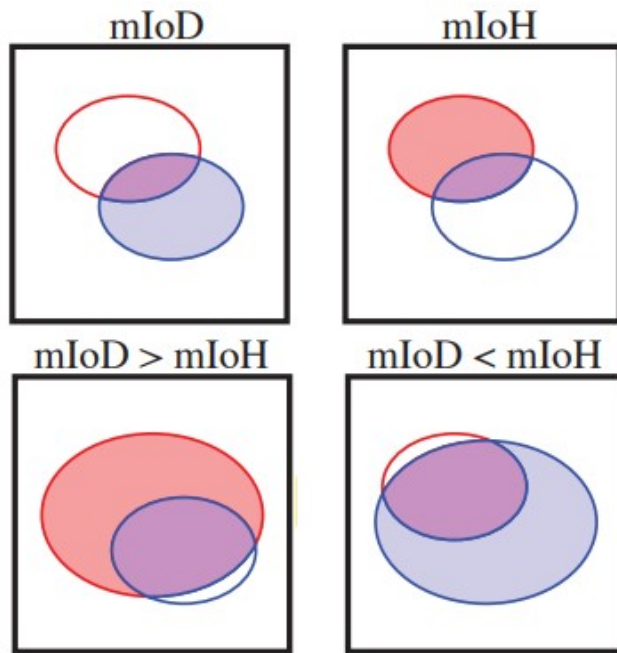
Head because main noun that represents the object being described

She ate the pizza

Head because main verb that describes the action being performed

Dependent because it is the noun that receives the action being performed

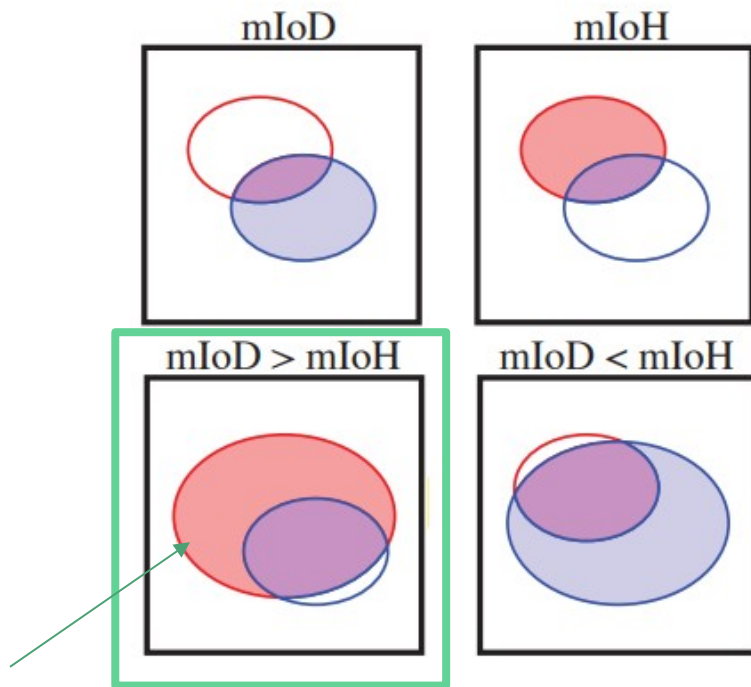
Visuosyntactic Analysis



Measures of Overlap

- Mean Intersection over Union (mIoU)
- Mean Intersection over the Dependent (mIoD)
- Mean Intersection over the Head (mIoH)

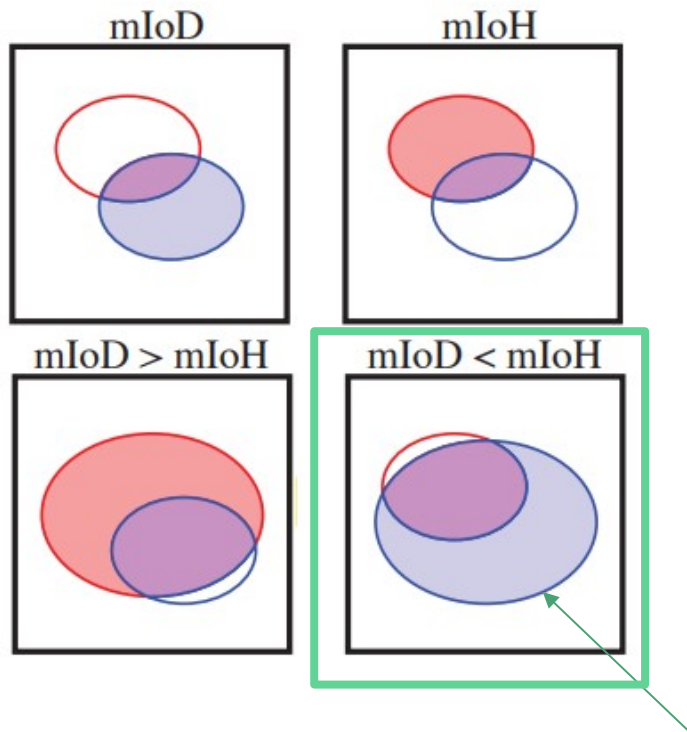
Visuosyntactic Analysis



Measures of Overlap

- Mean intersection over union (mIoU)
- Mean Intersection over the Dependent (mIoD)
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Visuosyntactic Analysis



Measures of Overlap

- Mean intersection over union (mIoU)
- Mean Intersection over the Dependent (mIoD)
- Mean Intersection over the Head (mIoH)

Visuosyntactic Analysis

#	Relation	mIoD	mIoH	Δ	mIoU
1	Unrelated pairs	65.1	66.1	1.0	47.5
2	All head-dependent pairs	62.3	62.0	0.3	43.4
3	compound	71.3	71.5	0.2	51.1
4	punct	68.2	70.0	1.8	49.5
5	nconj:and	58.0	56.1	1.9	38.2
6	det	54.8	52.2	2.6	35.0
7	case	51.7	58.1	6.4	36.9
8	acl	67.4	79.3	<u>12.</u>	55.4
9	nsubj	76.4	63.9	<u>12.</u>	52.2
10	amod	62.4	77.6	<u>15.</u>	51.1
11	nmod:of	73.5	57.9	<u>16.</u>	47.5
12	obj	75.6	46.3	<u>29.</u>	55.4
14	Coreferent word pairs	84.8	77.4	7.4	66.6

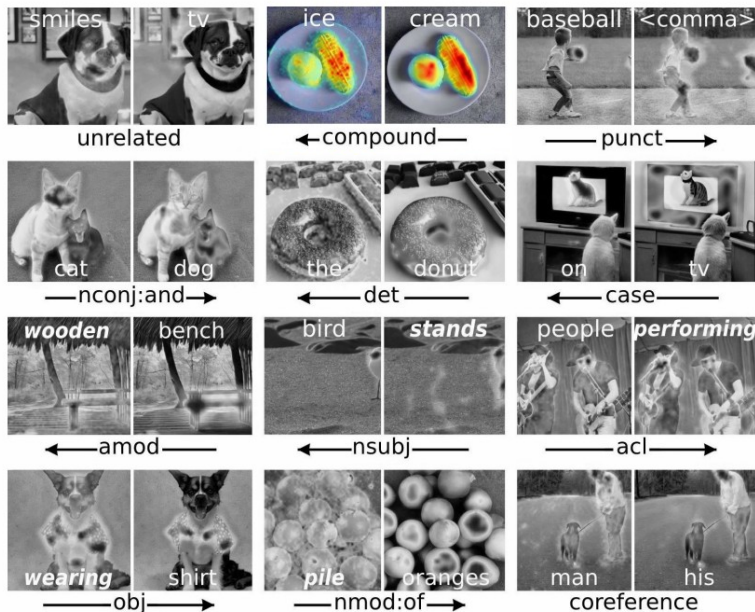
Noun Compounds

ice

+

cream

Visuosyntactic Analysis



Noun Compounds

ice

+

cream

Takeaway:

No dominance

Complement one another

Visuosyntactic Analysis

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Punctuation & Articles

the

+

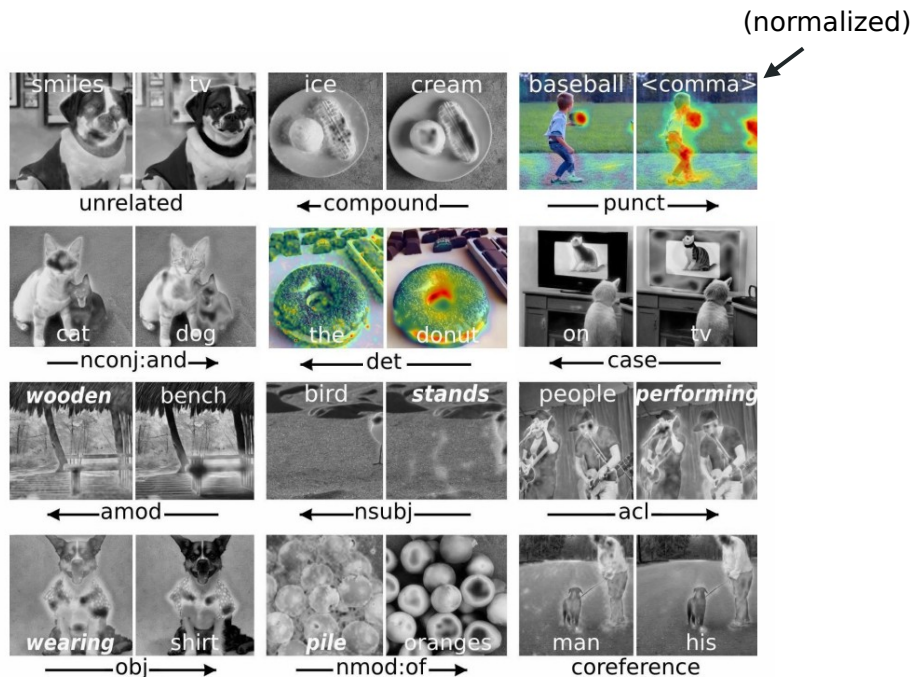
donut

baseball

+

<comma>

Visuosyntactic Analysis



Punctuation & Articles

the

+

donut

baseball

+

<comma>

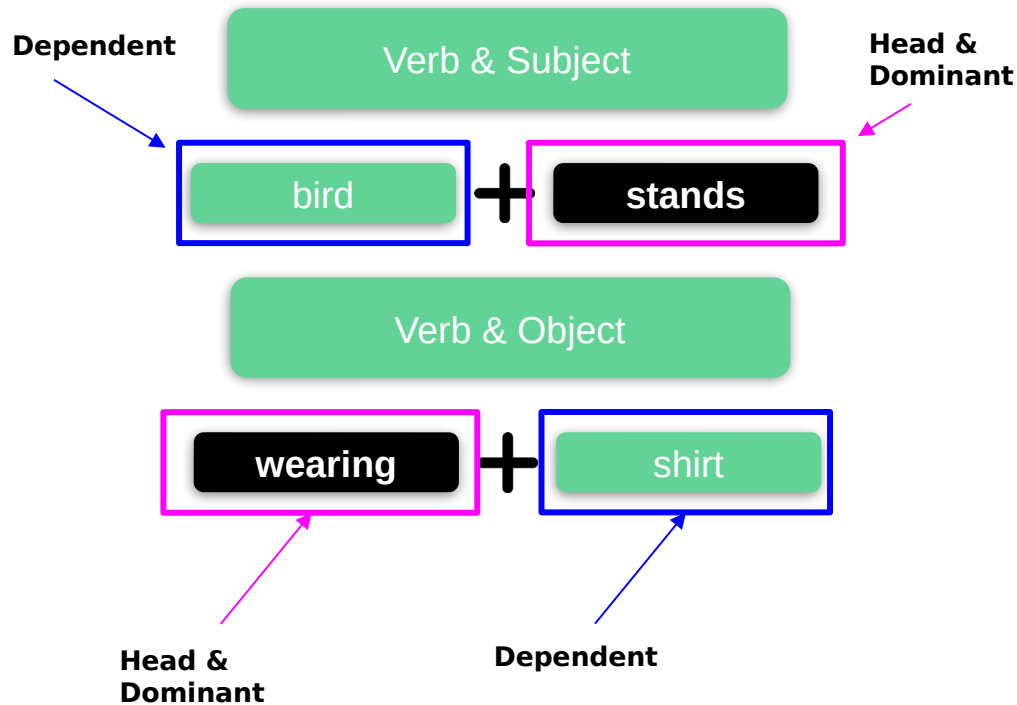
Takeaway:

No dominance

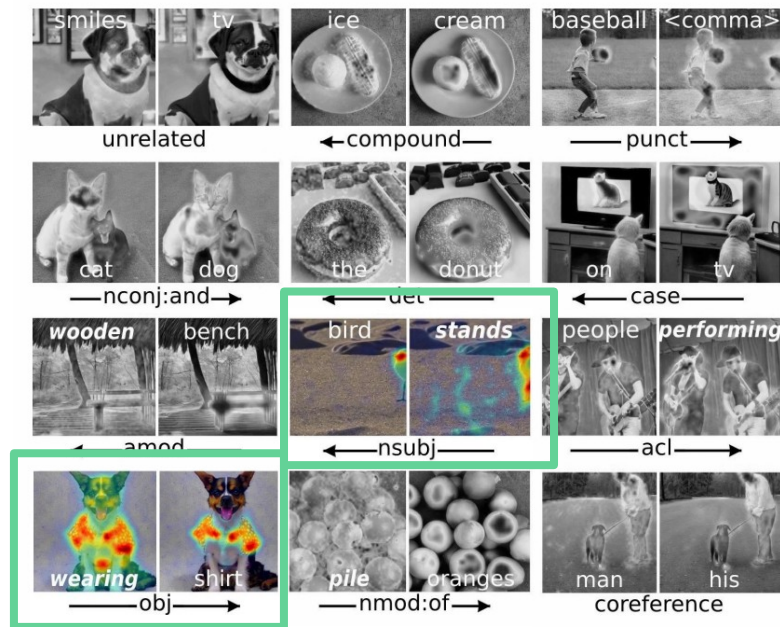
Little semantic meaning

Visuosyntactic Analysis

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Visuosyntactic Analysis



Dependent

Verb & Subject

Head & Dominant

bird

+

stands

Verb & Object

wearing

+

shirt

Head & Dominant

Takeaway:

Dependent

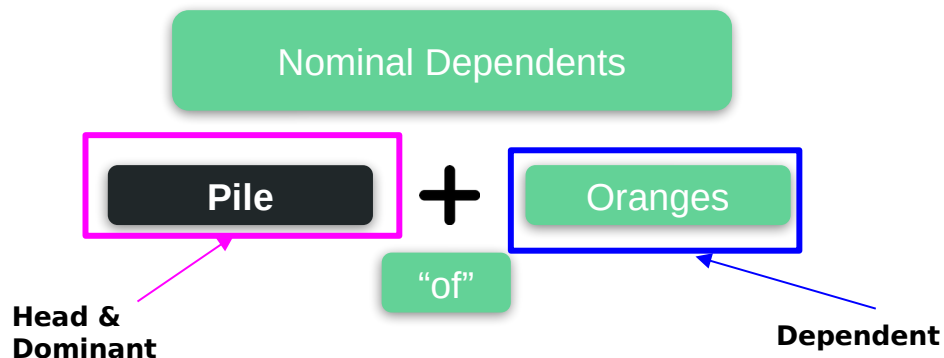
Semi-intuitive

Head verb dominates subject/object

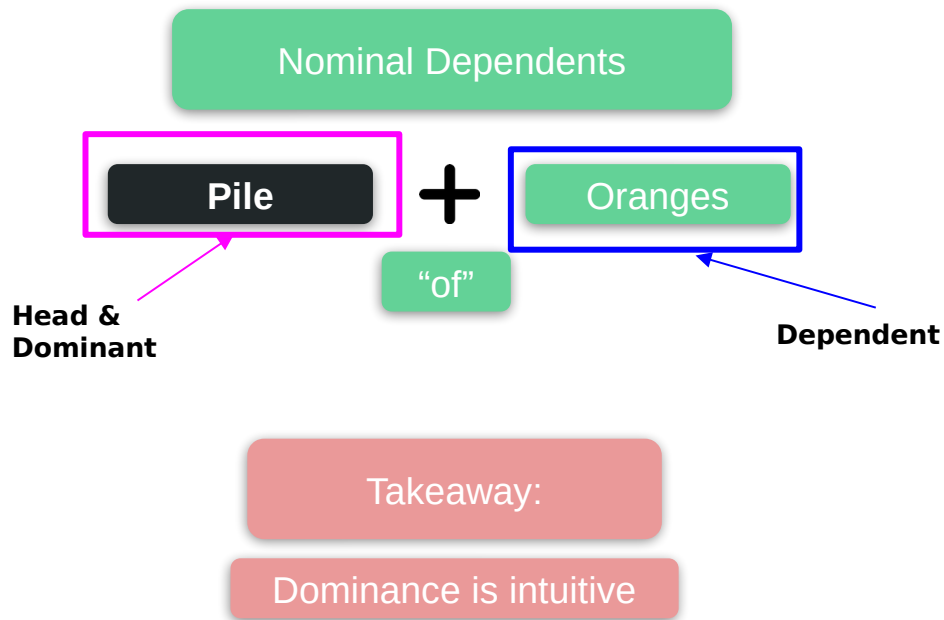
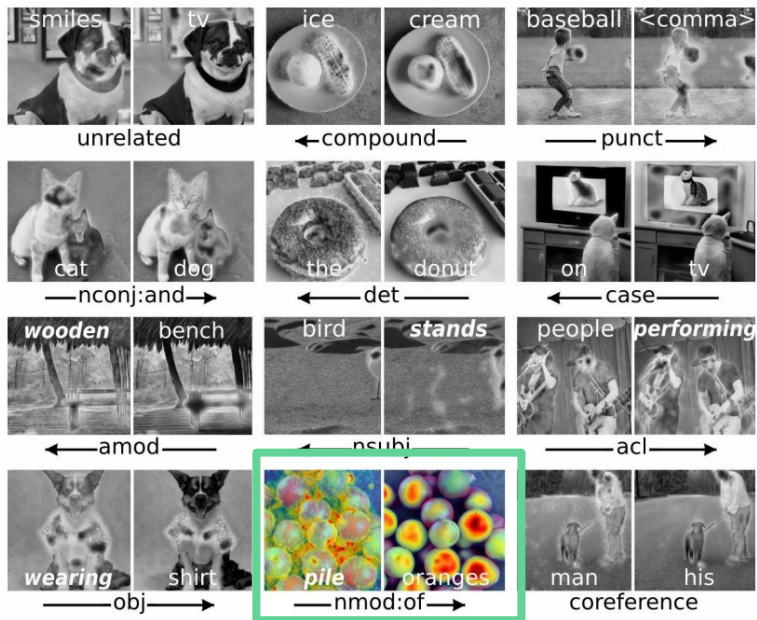
Verb contextualises the subject/object in its surroundings

Visuosyntactic Analysis

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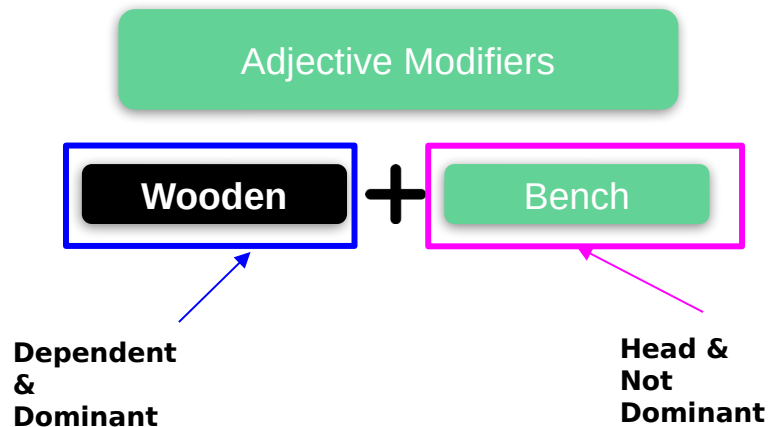


Visuosyntactic Analysis

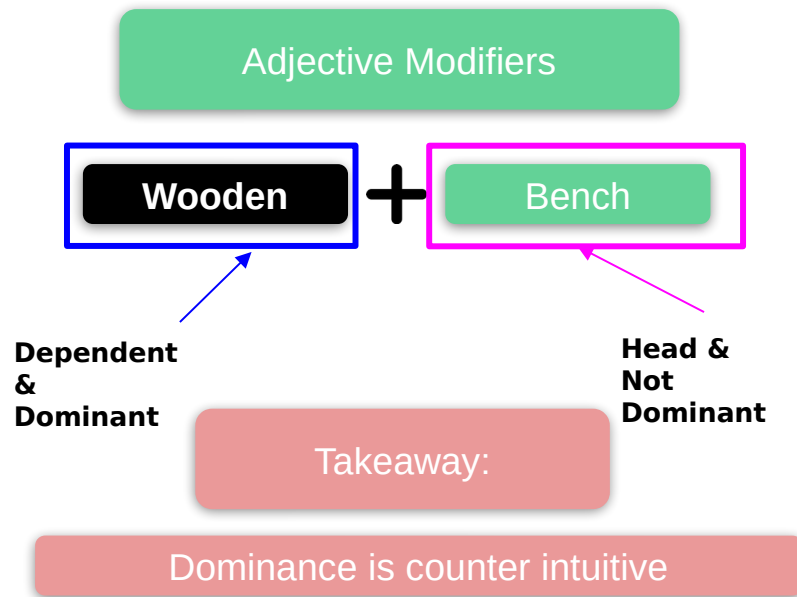
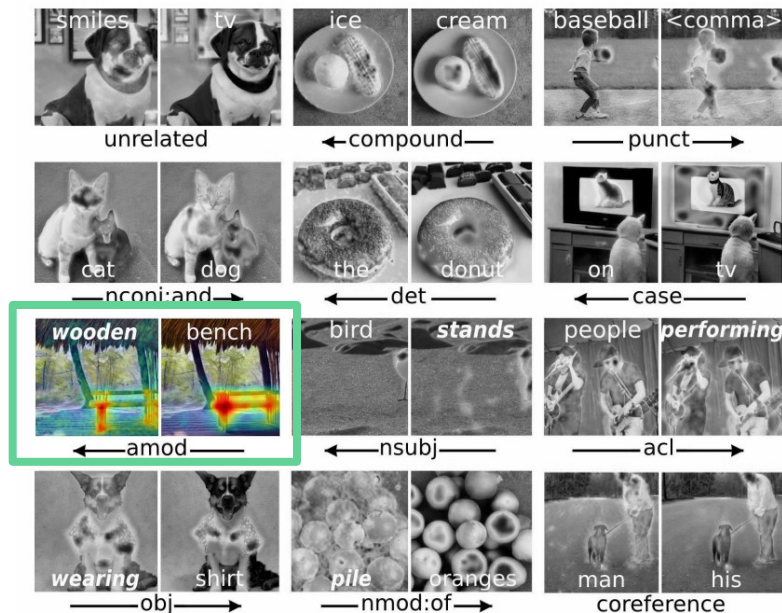


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Visuosyntactic Analysis



Visuosyntactic Analysis



How does **syntax** relate to the generated **pixels**?

Takeaway:

Attribution map of the **dependent** subsumes that of the **head**, and **opposite** for others

Dominance is **intuitive** in some cases but **counter intuitive** in others

Visuosemantic Analysis



Do semantically **similar words** have **worse** generation quality?

Visuosemantic Analysis: **Cohyponym Entanglement**

Prompt Structure:

“a(n) <noun> and a(n) <noun>”

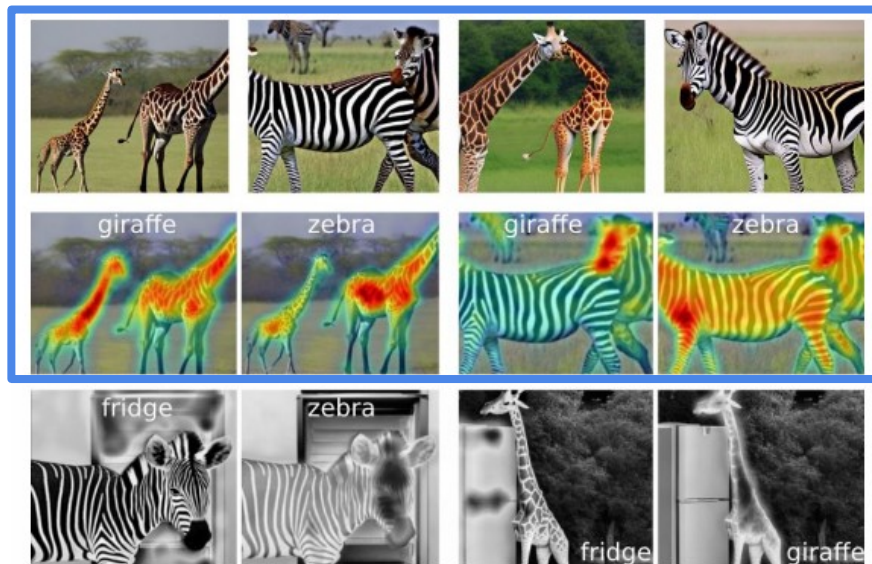
Cohyponym Example:

“a giraffe and a zebra”

Non-Cohyponym Example:

“a zebra and a fridge”

Visuosemantic Analysis: **Cohyponym Entanglement**



Cohyponym

“a giraffe and a zebra”

Takeaway

Stable-Diffusion image generation
worsens

Generates **one** of the nouns but **not both**

Attribution **maps** for the two nouns
overlap ... Feature Entanglement

Visuosemantic Analysis: **Cohyponym Entanglement**

Non-Cohyponym

“a zebra and a fridge”

Takeaway

Generates **both** of the nouns

Attribution **maps** for the two nouns
are **distinct**



Visuosemantic Analysis: **Adjectival Entanglement**

Prompt Structure:

“<adj> <noun> <verb phrase>”

Example:

“a [rusty] shovel sitting in a clean shed”

“a [bumpy] ball rolling down a hill”

Visuosemantic Analysis: **Adjectival Entanglement**

Expected Behavior

If **no entanglement**, background should **not gain attributes**
pertaining to that adjective

Visuosemantic Analysis: **Adjectival Entanglement**



Takeaway

Attribution maps for adjectives attend **too broadly** across images **beyond** nouns they modify ... **Feature Entanglement**

Summary

- DAAM provides pixel-level attribution maps for Stable Diffusion, a state-of-the-art text-to-image generator
- These maps appear to be informative, as evaluated through segmentation tasks and user study
- DAAM can be a useful tool for further understanding and analyzing Stable Diffusion – e.g. through visuosyntactic analysis and visuosemantic analysis

Class Discussion

- Does DAAM give a clear understanding about how a large-scale latent diffusion model synthesizes text to image and which parts of an image is influenced the most?
- Does DAAM explain all the dynamics of how images are synthesized? If not, how should DAAM be modified to better explain image generation?
 - Other explanatory tools besides **attention** and **segmentation proposals**?
- DAAM pointed out failure cases of stable-diffusion. Are there further interpretability methods needed to understand why **feature entanglement** is occurring and how it could be improved?